

HSIN-CHE WU

hcwu0975@iastate.edu | hsinchewu0975@gmail.com

2164 Edenburn Drive, Unit A, Ames IA 50010

Cell: 515-916-8831

Skill Set

Analog and Mixed Signal Circuits, Full Custom IC Layout, Spiking Neural Networks, Cadence Virtuoso, Xilinx Vivado, Verilog, Python, MATLAB, TSMC 180 BCD process, UMC 180 process.

Education

1. Ph.D. student in Electrical and Computer Engineering, Iowa State University, Jan. 2024 — now.
2. Master of Electrical Engineering, National Sun Yat-sen University, Taiwan, Sep. 2020—Nov. 2022
3. Bachelor of Electrical Engineering, National Sun Yat-sen University, Taiwan, Sep. 2016—Jun. 2020

Relevant Coursework (Iowa State University)

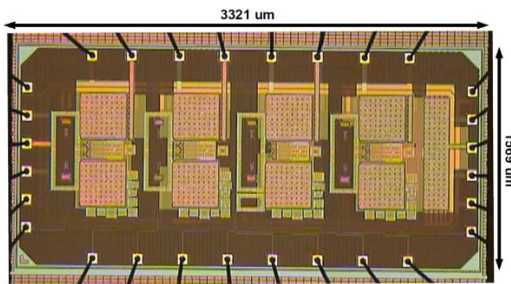
EE501 – Analog and Mixed-Signal VLSI Circuit Design Techniques (Grade: A)

EE503 – Power Management Integrated Circuits (Grade: A)

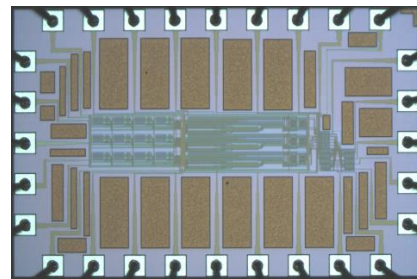
EE508 – Filter Design and Applications (Grade: A)

CPRE587 – Hardware Design for Machine Learning (Grade: A)

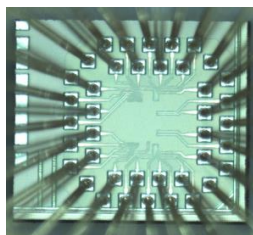
Tape-out Experience (Die photo attached)



4-Stage Negative Voltage Charge Pump
(TSMC 180 nm BCD)



8-bit Current Steering DAC
(UMC 180 nm)



8-bit Booth Multiplier (UMC 180 nm)

Patent (Taiwan)

C. -C. Wang, **H. -C. Wu** and T. -H. Lin, Negative Charge Pump System, Publication date: Feb. 21, 2023, publication number: I794123

Publications

1. C. -C. Wang, **H. -C. Wu** and T. -H. Lin, “A 4-Stage Negative Voltage Charge Pump with Randomly Selectable Parallel Switches,” *2022 IEEE 4th International Conference on Smart Power & Internet Energy Systems (SPIES)*, pp. 1502-1505, Dec. 2022.
2. C. -C. Wang, L. K. S. Tolentino, **H. -C. Wu**, R. G. B. Sangalang, O. L. J. A. Jose and T. -H. Lin, “A Negative Voltage Generator with 4-Stage Configurable Parallel Switching for Smart Window Film Applications,” *International Journal of Electronics Letters*, vol. 11, no. 3, pp. 361-379, May 2023.

3. C. -C. Wang, R. G. B. Sangalang, C.-P. Kuo, **H. -C. Wu**, Y. Hsu, S.-F. Hsiao, and C. -H. Yeh, "A 40.96-GOPS 196.8-mW Digital Logic Accelerator Used in DNN for Underwater Object Recognition," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 69, no. 12, pp. 4860-4871, Dec. 2022.
4. R. G. B. Sangalang, S. -H. Luo, **H. -C. Wu**, B. -Q. He, S. -F. Hsiao, C. -C. Wang, C. Jou, H. Hsia, D. C. -H. Yu, "A Power Effective DLA for PBs in Opto-Electrical Neural Network Architecture," *2022 IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)*, pp. 46-49, Nov. 2022.

Research Experience

1. Piezoelectric energy harvesting circuit – **National Science Foundation (NSF) project** (on going).
2. Proposed a wide-output negative charge pump circuit with 4-stage randomly configurable parallel switching architecture using TSMC 180-nm BCD Technology. The chip measurement result demonstrates the highest -14V voltage output at 0.5 mA current loading, and the correct function of enabling randomly negative charge pump stages, generating different negative output voltages. Furthermore, re-design the circuit base on measurement results, and achieve up to 77.66% power efficiency (simulation).
3. Proposes an optical convolution neural network system and a FPGA testing platform for MNIST handwritten number recognition in **Taiwan Semiconductor Manufacturing Co. (TSMC) Joint Developed Project (JDP)**.

Project Participation

1. (United States) National Science Foundation (NSF) - NSF-MeitY: Fundamental Realizable Limit Vibration Energy Harvesting: Design, Analysis, and Implementation (#2414972) (On going).
2. (Taiwan) National Science and Technology Council (NSTC) Moon-shot Project - Key Technology Development for Autonomous Underwater Vehicle with Artificial Intelligence (Sep. 2020 ~ Jul. 2022 project closed).
3. (Taiwan) Taiwan Semiconductor Manufacturing Company (TSMC) Joint Developed Project (JDP) - Digital Logic Accelerator Design Used in DNN/CNN for Object Recognition (Mar. 2022 ~ Dec. 2022).

Taiwan Semiconductor Research Institute (TSRI) Training Course

1. Chip Design Course: Cell-Based IC Physical Design and Verification with Innovus: 21 hours.
2. Chip Design Course: Edge-AI Realization with NeuroPilot – MediaTek AI Platform: 7 hours.

References

Ph.D. degree @ ISU:

Dr. Ratnesh Kumsr, Palmer Professor in Electrical and Computer Engineering at Iowa State University, Ames, Iowa.

E-mail: rkumar@iastate.edu